Changyeon Jo

System Software Engineer
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INDUSTRY EXPERIENCES

MangoBoost Inc., Seoul, South Korea — Nov 2022 - Present
System Software Engineer
Lead a team of 10 engineers performing system-level performance evaluation and validation of DPU products, focusing on network scalability and throughput. Led the company's MLPerf Storage benchmark submissions (v1.0, v2.0), achieving 6.2x higher GPU scalability compared to competing solutions. Developed the host device driver stack for an in-house RoCEv2 IP and built system-level validation for it in integrated environments.

Samsung Electronics, Hwaseong-si, South Korea — Sep 2021 - Oct 2022
Staff Engineer, Memory Software Development Team
Led the architecture design and implementation of a cloud data pipeline that ingests, cleans, and processes real-time SSD diagnostic data collected globally, and developed failure-pattern analyses and prediction models over approximately 10 years of accumulated data.

WorldQuant, Seoul, South Korea — Jun 2019 - Aug 2019
Research Consultant
Developed stock price prediction models using WorldQuant's internal alpha research tools.

EDUCATION

Seoul National University, Seoul, South Korea
Ph.D. in Computer Science — Mar 2012 - Aug 2021
Thesis: *Remote Memory for Virtualized Environments*
Advisor: Prof. Bernhard Egger

Hanyang University ERICA, Ansan, South Korea
B.S. in Computer Science — Mar 2008 - Feb 2012

RESEARCH EXPERIENCES

ETH Systems Group, Zürich, Switzerland — Mar 2018 - May 2018
Visiting Ph.D. Student
Advisor: Prof. Timothy Roscoe

PLASSE Lab, Hanyang University ERICA, South Korea — Jul 2010 - Jun 2011
Undergraduate Intern
Advisor: Prof. Kyoung-Goo Doh

RESEARCH PROJECTS

Instant Virtual Machine Live Migration — 2018 - 2021
Placing a virtual machine (VM) on a shared memory pool eliminates the memory copy that dominates live migration cost. We propose an optimized VM live migration technique for such RDMA-based remote memory environments, completing a migration in less than 100ms for memory-intensive workloads.

Remote Memory for Virtualized Environments — 2018 - 2020
Memory fragmentation across nodes leaves capacity unused, and a workload that does not fit in local memory falls back to disk. We built a software-based memory pool for virtualized environments, letting nodes share memory across an InfiniBand fabric via RDMA NICs. Our system reduced remote paging latency by 41.7x at the tail and improved job execution time by 3.5x under intensive remote paging scenarios.

Machine Learning Approach to Live Migration Modeling 2015 - 2017

Predicting the key performance metrics of VM live migration is notoriously difficult due to its complex behavior. We proposed a machine learning approach to live migration modeling; trained on 40,000 migration records, the model showed 2 to 5 times better prediction accuracy than the state-of-the-art analytical model. The resulting paper has been cited over 100 times.

Project page: https://csap.snu.ac.kr/software/lmdataset

VM Checkpoint, Restoration and Live Migration Techniques 2012 - 2015

We proposed fast and space efficient techniques for VM checkpoint, restoration, and live migration, reducing VM management overheads by 30% on average in the evaluation with real applications. The resulting paper has been cited over 100 times.

Project page: https://csap.snu.ac.kr/software/xencheckpointing

PUBLICATIONS

Kanghyun Choi, Bogyeong Park, Gihwan Lee, Hanmin Kim, Nayeon Kim, Chanmyeong Kim, Hyeonseong Choi, Dongjoo Lee, Jaehyun Lee, Eunjin Baek, Changsu Kim, **Changyeon Jo**, Minho Chu, and Jangwoo Kim. "MangoBoost FORMULA: A Fast, Scalable and Flexible AI RNIC." *In the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO'26), Industry Track*, Athens, Greece, 2026. To appear.

Youngsu Cho, **Changyeon Jo**, Reza Entezari-Maleki, Jörn Altmann, and Bernhard Egger. "Towards Smarter Live Migration: Minimizing SLO Violations and Costs." In *Future Generation Computer Systems, Volume 175, 108085*, 2026.

Heetaek Jeong, Wonsik Lee, Eunjin Baek, Changsu Kim, **Changyeon Jo**, Dongju Chae, Kanghyun Choi, Hamin Jang, Mohamed A. Elgammal, Sungmin Hong, Eriko Nurvitadhi, Dongup Kwon, and Jangwoo Kim. "MangoBoost Alice: Extremely Fast, Seamless, and Versatile FPGA-Accelerated DPU Solutions." In *IEEE Micro, Volume 45, Issue 5, pp. 43 - 55*, 2025.

Younghyun Cho, Jiyeon Park, Florian Negele, **Changyeon Jo**, Thomas R. Gross, and Bernhard Egger. "Dopia: Online Parallelism Management for Integrated CPU/GPU Architectures." *In 27th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP'22)*, April 2-6, 2022, Seoul, Republic of Korea.

Hyunik Kim, **Changyeon Jo**, and Bernhard Egger. "RapidSwap: A Hierarchical Far Memory." *Presented at the 18th International Conference on the Economics of Grids, Clouds, Systems and Services (GECON'21)*, Virtual Event, September 2021. In Lecture Notes in Computer Science (LNCS), Volume 13072, December 2021.

Daon Park, Hyeonsoo Kim, **Changyeon Jo**, and Bernhard Egger. "Can VM Live Migration Improve Job Throughput? Evidence from a Real World Cluster Trace" *Presented at the 18th International Conference on the Economics of Grids, Clouds, Systems and Services (GECON'21)*, Virtual Event, September 2021. In Lecture Notes in Computer Science (LNCS), Volume 13072, December 2021.

Changyeon Jo, Hyunik Kim, Hexiang Geng, and Bernhard Egger. "RackMem: A Tailored Caching Layer for Rack Scale Computing." *In Proceedings of the 29th International Conference on Parallel Architectures and Compilation Techniques (PACT'20)*, Virtual Event, October 2020.

Changyeon Jo, Hyunik Kim, and Bernhard Egger. "Instant Virtual Machine Live Migration." *In Proceedings of the 17th International Conference on the Economics of Grids Clouds, Systems and Services (GECON'20)*, Virtual Event, September 2020.

Youngsu Cho, **Changyeon Jo**, Hyunik Kim, and Bernhard Egger. "Towards Economical Live Migration in Data Centers." *In Proceedings of the 17th International Conference on the Economics of Grids Clouds, Systems and Services (GECON'20)*, Virtual Event, September 2020.

Changyeon Jo, Youngsu Cho, and Bernhard Egger. "A Machine Learning Approach to Live Migration Modeling." *In Proceedings of the 2017 ACM Symposium on Cloud Computing (SoCC'17)*, Santa Clara, USA, September 2017.

Changyeon Jo, Changmin Ahn, and Bernhard Egger. "A Machine Learning-based Approach to Live Migration Modeling." Presented at the *4th International Workshop on Efficient Data Center Systems (EDCS'16) co-located with ISCA'16*, Seoul, Korea, June 2016.

Bernhard Egger, Eunbyung Park, Younghyun Cho, **Changyeon Jo**, and Jaejin Lee. "Efficient Checkpointing of Live Virtual Machines." In *IEEE Transactions on Computers (TC), Volume 65, Issue 10, pp. 3041 - 3054*, January 2016.

Bernhard Egger, Erik Gustafsson, **Changyeon Jo**, and Jeongseok Son. "Efficiently restoring virtual machines." Presented at the *IFIP International Conference on Network and Parallel Computing (NPC'2013)*, Guiyang, China, September 2013, in *Springer International Journal of Parallel Programming (IJPP), Volume 43, Issue 3*, June 2015.

Changyeon Jo and Bernhard Egger. "Optimizing Live Migration for Virtual Desktop Clouds." In *Proceedings of the IEEE International Conference on Cloud Computing Technology and Science (CloudCom'2013)*, Bristol, UK, December 2013.

Changyeon Jo, Erik Gustafsson, Jeongseok Son, and Bernhard Egger. "Efficient live migration of virtual machines using shared storage." In *Proceedings of the ACM SIGPLAN/SIGOPS International Conference on Virtual Execution Environments (VEE'13)*, Houston, USA, March 2013.

Seonghun Jeong , Youngchul Cho, Daeyong Shin, **Changyeon Jo**, Yenjo Han, Soojung Ryu, Jeongwook Kim, and Bernhard Egger. "Random Test Program Generation for Reconfigurable Architectures." In *13th International Workshop on Microprocessor Test and Verification (MTV)*, Austin, USA, December 2012.

GRANTS

Young Researchers Exchange Program between Korea and Switzerland, Swiss State Secretariat for Education, Research and Innovation (SERI), 2018

ACM SIGMOD Travel Grants, ACM Symposium on Cloud Computing, 2017

TEACHING EXPERIENCES

Teaching Assistant, Seoul National University 2012 - 2018
Eight semesters of System Programming (4190.203, M1522.000800) and Computer Architecture (4190.308).

PROFESSIONAL SERVICES

Artifact Evaluation Committee, *International Conference on Languages, Compilers, Tools and Theory of Embedded Systems (LCTES)*, 2019

External Reviewer, *IEEE Transactions on Cloud Computing*, 2015

DOMAINS

RDMA networking and congestion control, collective communication libraries, memory and storage disaggregation, DPU and SmartNIC software stacks, Linux kernel, SPDK, virtualization, system performance engineering and benchmarking, machine learning for systems